

## RaceDay API Endpoint Plan

**Module:** PROG6212 - Programming 2B

Part 1 - System Planning and Database

| HTTP Method | Route              | Description                                           | Role Required | Request Body                                        | Expected Response                                                                                                    |
|-------------|--------------------|-------------------------------------------------------|---------------|-----------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| <b>POST</b> | /api/auth/register | Creates a new user account (Organiser or Participant) | None          | {firstName, lastName, email, password, role, phone} | 201 Created – user created + token<br>400 Bad Request – validation failed<br>409 Conflict – email already registered |
| <b>POST</b> | /api/auth/login    | Authenticates a user and returns an access token.     | None          | {email, password}                                   | 200 OK – token + user info<br>401 Unauthorized – invalid credentials                                                 |
| <b>GET</b>  | /api/users/me      | Returns the profile of the currently logged-in user.  | Any           | None                                                | 200 OK – user profile<br>401 Unauthorized                                                                            |
| <b>PUT</b>  | /api/users/me      | Updates the profile of the currently logged-in user.  | Any           | {firstName, lastName, phone}                        | 200 OK – profile updated<br>400 Bad Request                                                                          |
| <b>GET</b>  | /api/events        | Lists all upcoming events.                            | None          | None                                                | 200 OK – list of events                                                                                              |
| <b>GET</b>  | /api/events/{id}   | Returns full details for a                            | None          | None                                                | 200 OK – event                                                                                                       |

|               |                                         |                                                 |                   |                                                          |                                                            |
|---------------|-----------------------------------------|-------------------------------------------------|-------------------|----------------------------------------------------------|------------------------------------------------------------|
|               |                                         | specific event.                                 |                   |                                                          | details 404 not found                                      |
| <b>POST</b>   | /api/events                             | Creates a new event.                            | Organiser         | {eventName, description, eventDate, location, eventType} | 201 Created – event created 400 Bad Request                |
| <b>PUT</b>    | /api/events/{id}                        | Updates an existing event.                      | Organiser (Owner) | {eventName, description, eventDate, location, eventType} | 200 OK – event updated 403 Forbidden 404 Not Found         |
| <b>DELETE</b> | /api/events/{id}                        | Deletes an event.                               | Organiser (Owner) | None                                                     | 200 OK – deleted 403 Forbidden 404 Not Found               |
| <b>GET</b>    | /api/events/{eventId}/categories        | Lists all categories for a specific event.      | None              | None                                                     | 200 OK – list of categories 404 Not Found                  |
| <b>POST</b>   | /api/events/{eventId}/categories        | Adds a new category to an event.                | Owner (Owner)     | {categoryName, description, distance, entryFee}          | 201 Created – category created 403 Forbidden 404 Not Found |
| <b>PUT</b>    | /api/categories/{id}                    | Updates an existing category.                   | Organiser (Owner) | {categoryName, description, distance, entryFee}          | 200 OK – category updated 403 Forbidden 404 Not Found      |
| <b>DELETE</b> | /api/categories/{id}                    | Deletes a category.                             | Organiser (Owner) | None                                                     | 200 OK – deleted 403 Forbidden 404 Not Found               |
| <b>POST</b>   | /api/categories/{categoryId}/enrolments | Enrols the logged-in participant in a category. | Participant       | { }<br>(Participant Id taken)                            | 201 Created – enrolment created 404 Not Found              |

|               |                                       |                                                                           |                         |                                 |                                                                                                           |
|---------------|---------------------------------------|---------------------------------------------------------------------------|-------------------------|---------------------------------|-----------------------------------------------------------------------------------------------------------|
|               |                                       |                                                                           |                         | <b>from token)</b>              | 409 Conflict already enrolled/ category full                                                              |
| <b>GET</b>    | /api/users/me/enrolments              | Lists the enrolments of the logged-in participant.                        | Participant             | <b>None</b>                     | 200 OK i list of enrolments                                                                               |
| <b>GET</b>    | /api/events/{eventId}/enrolments      | Lists all enrolments for a specific events.                               | Organiser (owner)       | <b>None</b>                     | 200 OK – of enrolments<br>403 Forbidden                                                                   |
| <b>DELETE</b> | /api/enrolments/{id}                  | Cancels an enrolment.                                                     | Participant (owner)     | <b>None</b>                     | 200 OK – cancelled<br>403 Forbidden                                                                       |
| <b>POST</b>   | /api/enrolments/{enrolmentId}/results | Captures a participant's result.                                          | Organiser (event owner) | <b>{ finishTime, position }</b> | 201 Created – result recorded<br>403 Forbidden<br>404 Not found<br>409 Conflict – result already captured |
| <b>GET</b>    | /api/users/me/results                 | Returns the performance history of the logged-in participant              | Participant             | <b>None</b>                     | 200 OK – list of results                                                                                  |
| <b>GET</b>    | /api/events/{eventId}/results         | Returns full results for a specific event.                                | None                    | <b>None</b>                     | 200 OK - list of results<br>404 Not Found                                                                 |
| <b>GET</b>    | /api/events/{eventId}/route           | Returns the route information of an event (weather, distance, departure/a | None                    | <b>None</b>                     | 200 OK — of route information<br>•404 Not Found                                                           |

|             |                             |                                          |                              |                                                                      |                                                          |
|-------------|-----------------------------|------------------------------------------|------------------------------|----------------------------------------------------------------------|----------------------------------------------------------|
|             |                             | rival<br>points)                         |                              |                                                                      |                                                          |
| <b>POST</b> | /api/events/{eventId}/route | Adds route<br>information<br>to an event | Organi<br>ser<br>(Owner<br>) | <b>{routeName,<br/>description,<br/>startLocation,<br/>distance}</b> | 201<br>Created•40<br>3<br>Forbidden•<br>404 Not<br>Found |
| <b>PUT</b>  | /api/events/{eventId}/route | Change the<br>course<br>information      | Organi<br>ser<br>(Owner<br>) | <b>{routeName,<br/>description,<br/>startLocation,<br/>distance}</b> | 200 OK•<br>403<br>Forbidden•<br>404 Not<br>Found         |